

This exam contains two questions. You may consult your class notes, my lecture notes, handouts and any one quantum field theory textbook. In order to receive proper credit for your answers you must show how you arrived at them.

1. (10 points) Consider a field theory of a neutral pseudoscalar meson (field operator π , mass m) and a neutral scalar (field operator ϕ , mass λ) interacting with fermions (field operator ψ , mass M) The interaction lagrangian is,

$$L = -g_1 \partial_\mu \phi \bar{\psi} \gamma^\mu \psi - g_2 \pi \bar{\psi} i \gamma_5 \psi + g (\partial_\mu \phi)^2 \pi$$

- (i) What are the mass dimensions of the coupling constants g , g_1 and g_2 ?
 - (ii) What are the Feynman rules for this theory?
 - (iii) Draw all relevant Feynman diagrams and write down the corresponding S matrix elements for the processes:
 - (a) $\phi(k_1) + \pi(k_2) \rightarrow \psi(p_1) + \bar{\psi}(p_2)$
 - (b) $\phi(k_1) + \phi(k_2) \rightarrow \psi(p_1) + \bar{\psi}(p_2)$
2. (10 points) Consider a theory of a pseudoscalar meson (π , mass m) coupled to fermions (ψ , mass M) and also to a scalar meson (ϕ , mass λ). The interaction lagrangian is:

$$L = -g_2 \pi \bar{\psi} i \gamma_5 \psi + g (\partial_\mu \phi)^2 \pi$$

- (a) To lowest nontrivial order in the coupling constants g and g_2 write down the S-matrix element for the process $\psi \bar{\psi} \rightarrow \phi \phi$.
- (b) Calculate the differential cross-section $\frac{d\sigma}{d\Omega}$ in the CM frame..

Physics 513 Final

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My notation mostly follows Schwartz.

1. 1

The interaction Lagrangian is

$$\mathcal{L}_1 = g(\partial\phi)^2\pi - g_1(\partial\phi)\bar{\psi}\gamma^\mu\psi - g_2\pi\bar{\psi}i\gamma_5\psi. \quad (1)$$

1.1. (i) Units

I work in units where $\hbar = c = 1$. Then everything is measured in powers of energy E . So $[\mathcal{O}] = E^p$, or more succinctly $[\mathcal{O}] = p$. The action is

$$S = \int d^4x \mathcal{L}. \quad (2)$$

The units in this equation are

$$[S] = 0, \quad [x] = -1, \quad [d^4x] = -4, \quad [\mathcal{L}] = 4. \quad (3)$$

From Schwartz Eq. (A.9) scalar fields have units $[\phi] = 1$. I assume the same holds for pseudoscalar fields $[\pi] = 1$.

$$[g(\partial\phi)^2\pi] = 4 \quad (4)$$

$$[g(\partial\phi)^2] + [\pi] = 4 \quad (5)$$

$$[g(\partial\phi)^2] + 1 = 4 \quad (6)$$

$$[g(\partial\phi)^2] = 3 \quad (7)$$

$$[g] + 2[\partial\phi] = 3 \quad (8)$$

$$[\partial\phi] = \left[\frac{d\phi}{dx}\right] = \left[\frac{\phi}{x}\right] = [\phi] - [x] = 1 + 1 = 2 \quad (9)$$

$$[g] + 4 = 3 \quad (10)$$

$$\boxed{[g] = -1} \quad (11)$$

$$[g_1(\partial\phi)\bar{\psi}\gamma^\mu\psi] = 4 \quad (12)$$

$$[g_1\bar{\psi}\gamma^\mu\psi] + 2 = 4 \quad (13)$$

$$[g_1\bar{\psi}\gamma^\mu\psi] = 2 \quad (14)$$

$$[g_1] + [\bar{\psi}] + [\gamma] + [\psi] = 2 \quad (15)$$

Gamma matrices are collections of pure numbers and must be dimensionless: $[\gamma] = 0$

$$[g_1] + [\bar{\psi}] + [\psi] = 2 \quad (16)$$

$$\bar{\psi} = \psi^\dagger\gamma^0 \implies [\bar{\psi}] = [\psi] \quad (17)$$

$$[g_1] + 2[\psi] = 2 \quad (18)$$

The free Dirac Lagrangian has the term $\bar{\psi}M\psi$ so

$$[\bar{\psi}M\psi] = 4, \quad (19)$$

$$2[\psi] + [M] = 4, \quad (20)$$

$$2[\psi] + 1 = 4, \quad (21)$$

$$\boxed{[\psi] = 3/2}. \quad (22)$$

$$[g_1] + 3 = 2 \quad (23)$$

$$\boxed{[g_1] = -1} \quad (24)$$

$$[g_2\pi\bar{\psi}i\gamma_5\psi] = 4 \quad (25)$$

$$[g_2] + [\pi] + [\bar{\psi}] + [\cancel{i\gamma_5}] + [\psi] = 4 \quad (26)$$

$$[g_2] + [\pi] + 2[\psi] = 4 \quad (27)$$

$$[g_2] + 4 = 4 \quad (28)$$

$$\boxed{[g_2] = 0} \quad (29)$$

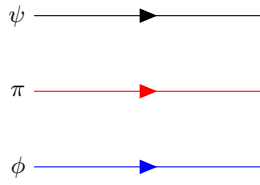
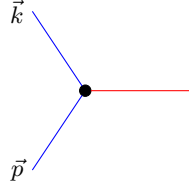


FIG. 1. Lines for different fields.

FIG. 2. Vertex for $g(\partial\phi)^2\pi$, with factor $-2ig(\pm\vec{p}) \cdot (\pm\vec{k})$, where the + is for leaving the vertex and the - is for entering.

1.2. (ii) Feynman Rules

I denote the three fields using the lines in Fig. 1. The propagators are

$$\begin{aligned}\psi &\sim i \frac{\not{p} + M}{\vec{p}^2 - M^2 + i\epsilon}, \\ \pi &\sim \frac{i}{\vec{p}^2 - m^2 + i\epsilon}, \\ \phi &\sim \frac{i}{\vec{p}^2 - \lambda^2 + i\epsilon}.\end{aligned}\quad (30)$$

The three terms in the Lagrangian yield the three interaction vertices in Figs. 2-4.

1.3. (iii) Feynman Diagrams & $\mathcal{M}$

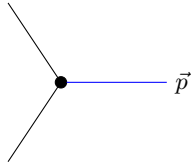
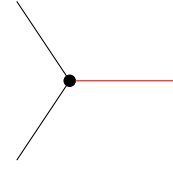
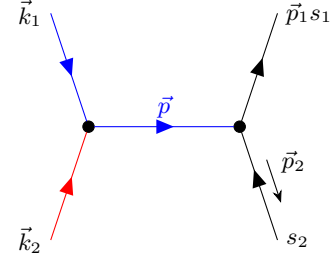
1.3.1. (a) $\phi\pi \rightarrow \psi\bar{\psi}$

The reaction $\phi\pi \rightarrow \psi\bar{\psi}$ is equivalent to $\phi\pi \rightarrow e^-e^+$ so I take

$$\vec{k}_1(\phi) + \vec{k}_2(\pi) \rightarrow \vec{p}_1(e^-) + \vec{p}_2(e^+). \quad (31)$$

The total momenta are

$$\vec{K} = \vec{k}_1 + \vec{k}_2, \quad \vec{P} = \vec{p}_1 + \vec{p}_2. \quad (32)$$

FIG. 3. Vertex for $-g_1(\partial\phi)\bar{\psi}\gamma^\mu\psi$, with factor $g_1(\pm p_\mu)\gamma^\mu = g_1(\pm\not{p})$, where the + is for leaving the vertex and the - is for entering.FIG. 4. Vertex for $-g_2\pi\bar{\psi}i\gamma_5\psi$, with factor $g_2\gamma_5$.FIG. 5. Feynman diagram $\phi[1]$. I only add external momentum arrows for positrons.

Since this is 2 to 2 scattering I use Mandelstam variables

$$\begin{aligned}s &= (\vec{k}_1 + \vec{k}_2)^2 = (\vec{p}_1 + \vec{p}_2)^2, \\ t &= (\vec{k}_1 - \vec{p}_1)^2 = (\vec{k}_2 - \vec{p}_2)^2, \\ u &= (\vec{k}_1 - \vec{p}_2)^2 = (\vec{k}_2 - \vec{p}_1)^2.\end{aligned}\quad (33)$$

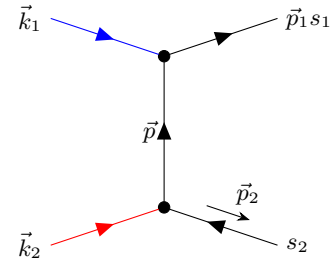
The non-trivial part of the S-matrix, $\mathcal{M}$, has 3 diagrams (Figs. 5 - 7) to lowest order.

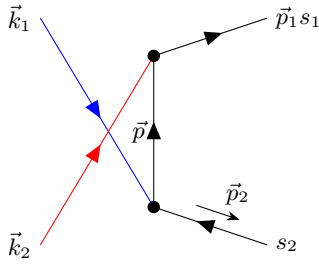
$$i\delta^{(4)}(\vec{K} - \vec{P})\mathcal{M} = \phi[1] + \phi[2] + \phi[3] \quad (34)$$

I use the shorthand $u_n = u_{s_n}(\vec{p}_n)$.

$$\begin{aligned}\phi[1] &= -2g \int d^4p \frac{\bar{u}_1 v_2 (\vec{p} \cdot \vec{k}_1)}{\vec{p}^2 - \lambda^2 + i\epsilon} \\ &\quad \delta^{(4)}(\vec{k}_1 + \vec{k}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_1 - \vec{p}_2)\end{aligned}\quad (35)$$

$$\phi[1] = -2g \int d^4p \frac{\bar{u}_1 v_2 (\vec{p} \cdot \vec{k}_1)}{\vec{p}^2 - \lambda^2 + i\epsilon} \delta^{(4)}(\vec{K} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}) \quad (36)$$

FIG. 6. $\phi[2]$

FIG. 7. $\mathfrak{Y}[3]$

$$\mathfrak{Y}[1] = -2g \frac{\bar{u}_1 v_2 (\vec{K} \cdot \vec{k}_1)}{\vec{K}^2 - \lambda^2} \delta^{(4)}(\vec{K} - \vec{P}) \quad (37)$$

$$s = (\vec{k}_1 + \vec{k}_2)^2 = \vec{K}^2 \quad (38)$$

$$\vec{K} \cdot \vec{k}_1 = \vec{k}_1^2 + \vec{k}_2 \cdot \vec{k}_1 = \lambda^2 + \vec{k}_2 \cdot \vec{k}_1 \quad (39)$$

$$\mathfrak{Y}[1] = -\delta^{(4)}(\vec{K} - \vec{P}) \frac{2g\bar{u}_1 v_2}{s - \lambda^2} (\lambda^2 + \vec{k}_2 \cdot \vec{k}_1) \quad (40)$$

$$\begin{aligned} \mathfrak{Y}[2] = & -ig_1 g_2 \int d^4 p \bar{u}_1 k_1 \frac{\not{p} + M}{\vec{p}^2 - M^2 + i\epsilon} \gamma_5 v_2 \\ & \delta^{(4)}(\vec{p} + \vec{k}_1 - \vec{p}_1) \delta^{(4)}(\vec{k}_2 - \vec{p}_2 - \vec{p}) \end{aligned} \quad (41)$$

Because $\vec{k}_1$ is associated with the $\vec{p}_1$ electron vertex in $\mathfrak{Y}[2]$ I place k_1 with $\bar{u}_1$. The order $\bar{u}_1 k_1$ is required to yield a scalar.

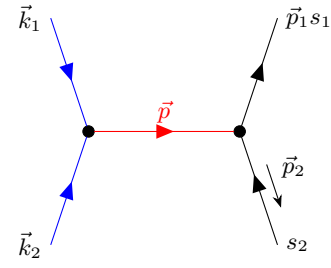
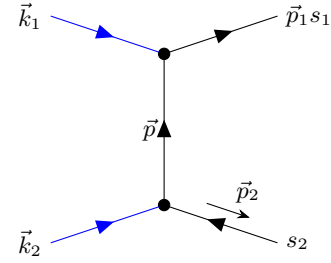
$$\begin{aligned} \mathfrak{Y}[2] = & -ig_1 g_2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{(\vec{k}_2 - \vec{p}_2)^2 - M^2} \gamma_5 v_2 \\ & \delta^{(4)}(\vec{k}_1 + \vec{k}_2 - \vec{p}_1 - \vec{p}_2) \end{aligned} \quad (42)$$

$$\mathfrak{Y}[2] = -i\delta^{(4)}(\vec{K} - \vec{P}) g_1 g_2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{(\vec{k}_2 - \vec{p}_2)^2 - M^2} \gamma_5 v_2 \quad (43)$$

$$t = (\vec{k}_2 - \vec{p}_2)^2 \quad (44)$$

$$\mathfrak{Y}[2] = -i\delta^{(4)}(\vec{K} - \vec{P}) g_1 g_2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{t - M^2} \gamma_5 v_2 \quad (45)$$

$$\begin{aligned} \mathfrak{Y}[3] = & -ig_1 g_2 \int d^4 p \bar{u}_1 \gamma_5 \frac{\not{p} + M}{\vec{p}^2 - M^2 + i\epsilon} k_1 v_2 \\ & \delta^{(4)}(\vec{p} + \vec{k}_2 - \vec{p}_1) \delta^{(4)}(\vec{k}_1 - \vec{p}_2 - \vec{p}) \end{aligned} \quad (46)$$

FIG. 8. $\mathfrak{Y}[1]$ FIG. 9. $\mathfrak{Y}[2]$

$$\mathfrak{Y}[3] = -i\delta^{(4)}(\vec{K} - \vec{P}) g_1 g_2 \bar{u}_1 \gamma_5 \frac{k_1 - \not{p}_2 + M}{(\vec{k}_1 - \vec{p}_2)^2 - M^2} k_1 v_2 \quad (47)$$

$$u = (\vec{k}_1 - \vec{p}_2)^2 \quad (48)$$

$$\mathfrak{Y}[3] = -i\delta^{(4)}(\vec{K} - \vec{P}) g_1 g_2 \bar{u}_1 \gamma_5 \frac{k_1 - \not{p}_2 + M}{u - M^2} k_1 v_2 \quad (49)$$

$$\begin{aligned} \mathcal{M} = & i \frac{2g\bar{u}_1 v_2}{s - \lambda^2} (\lambda^2 + \vec{k}_2 \cdot \vec{k}_1) \\ & - g_1 g_2 \bar{u}_1 \left(k_1 \frac{k_2 - \not{p}_2 + M}{t - M^2} \gamma_5 + \gamma_5 \frac{k_1 - \not{p}_2 + M}{u - M^2} k_1 \right) v_2 \end{aligned} \quad (50)$$

1.3.2. (b) $\phi\phi \rightarrow \psi\bar{\psi}$

$$\vec{k}_1(\phi) + \vec{k}_2(\phi) \rightarrow \vec{p}_1(e^-) + \vec{p}_2(e^+). \quad (51)$$

$$i\delta^{(4)}(\vec{K} - \vec{P}) \mathcal{M} = \mathfrak{Y}[1] + \mathfrak{Y}[2] \quad (52)$$

I show the Feynman diagrams in Figs. 8 & 9.

2.1. (a)

$$\mathfrak{V}[1] = 2gg_2 \int d^4p \frac{(\vec{k}_1 \cdot \vec{k}_2) \bar{u}_1 \gamma_5 v_2}{\vec{p}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{k}_1 + \vec{k}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_1 - \vec{p}_2) \quad (53)$$

$$\mathfrak{V}[1] = 2gg_2 \int d^4p \frac{(\vec{k}_1 \cdot \vec{k}_2) \bar{u}_1 \gamma_5 v_2}{\vec{p}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{K} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}) \quad (54)$$

$$\mathfrak{V}[1] = \delta^{(4)}(\vec{K} - \vec{P}) 2gg_2 \frac{(\vec{k}_1 \cdot \vec{k}_2) \bar{u}_1 \gamma_5 v_2}{\vec{K}^2 - m^2} \quad (55)$$

$$\mathfrak{V}[1] = \delta^{(4)}(\vec{K} - \vec{P}) 2gg_2 \frac{(\vec{k}_1 \cdot \vec{k}_2) \bar{u}_1 \gamma_5 v_2}{s - m^2} \quad (56)$$

$$\mathfrak{V}[2] = ig_1^2 \int d^4p \bar{u}_1 k_1 \frac{\not{p} + M}{\vec{p}^2 - M^2 + i\epsilon} k_2 v_2 \delta^{(4)}(\vec{p} + \vec{k}_1 - \vec{p}_1) \delta^{(4)}(\vec{k}_2 - \vec{p}_2 - \vec{p}) \quad (57)$$

$$\mathfrak{V}[2] = i\delta^{(4)}(\vec{K} - \vec{P}) g_1^2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{(\vec{k}_2 - \vec{p}_2)^2 - M^2} k_2 v_2 \quad (58)$$

$$\mathfrak{V}[2] = i\delta^{(4)}(\vec{K} - \vec{P}) g_1^2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{t - M^2} k_2 v_2 \quad (59)$$

$$\mathcal{M} = g_1^2 \bar{u}_1 k_1 \frac{k_2 - \not{p}_2 + M}{t - M^2} k_2 v_2 - 2igg_2 \frac{(\vec{k}_1 \cdot \vec{k}_2) \bar{u}_1 \gamma_5 v_2}{s - m^2} \quad (60)$$

$$\mathcal{M} = \bar{u}_1 \left(g_1^2 k_1 \frac{k_2 - \not{p}_2 + M}{t - M^2} k_2 - 2igg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \gamma_5 \right) v_2 \quad (61)$$

2. 2

The interaction Lagrangian is

$$\mathcal{L}_1 = g(\partial\phi)^2\pi - g_2\pi\bar{\psi}i\gamma_5\psi. \quad (62)$$

The Feynman rules for this Lagrangian are given in Sec. 1.2, except that $g_1 = 0$.

The reaction $\psi\bar{\psi} \rightarrow \phi\phi$ is equivalent to $e^-e^+ \rightarrow \phi\phi$.

$$\vec{p}_1(e^-) + \vec{p}_2(e^+) \rightarrow \vec{k}_1(\phi) + \vec{k}_2(\phi) \quad (63)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{k}_1 + \vec{k}_2 \quad (64)$$

I use the Mandelstam variables in Eq. 33.

By time reversal symmetry $\mathcal{M}$ is given by Eq. 61 (with $g_1 = 0$).

$$\mathcal{M} = -2igg_2 \frac{\bar{u}_1 \gamma_5 v_2}{s - m^2} (\vec{k}_1 \cdot \vec{k}_2) \quad (65)$$

Since only $|\mathcal{M}|^2$ is observable I drop the $-i$.

$$\mathcal{M} = 2gg_2 \frac{\bar{u}_1 \gamma_5 v_2}{s - m^2} (\vec{k}_1 \cdot \vec{k}_2) \quad (66)$$

2.2. (b)

2.2.1. $\mathcal{M}^\dagger$

$$\mathcal{M}^\dagger = 2gg_2 \frac{(\bar{u}_1 \gamma_5 v_2)^\dagger}{s - m^2} (\vec{k}_1 \cdot \vec{k}_2) \quad (67)$$

$$Q = (\bar{u} \gamma_5 v)^\dagger \quad (68)$$

$$\bar{u} = u^\dagger \gamma^0 \quad (69)$$

$$Q = (u^\dagger \gamma^0 \gamma_5 v)^\dagger = v^\dagger \gamma_5^\dagger \gamma^{0\dagger} u \quad (70)$$

$$\gamma^{0\dagger} = \gamma^0, \quad \gamma_5^\dagger = \gamma_5 \quad (71)$$

$$Q = v^\dagger \gamma_5 \gamma^0 u \quad (72)$$

$$\{\gamma_5, \gamma^\mu\} = 0 \implies \gamma_5 \gamma^\mu = -\gamma^\mu \gamma_5 \quad (73)$$

$$Q = -v^\dagger \gamma^0 \gamma_5 u = -\bar{v} \gamma_5 u \quad (74)$$

$$(\bar{u} \gamma_5 v)^\dagger = -\bar{v} \gamma_5 u \quad (75)$$

$$\mathcal{M}^\dagger = -2gg_2 \frac{\bar{v}_2 \gamma_5 u_1}{s - m^2} (\vec{k}_1 \cdot \vec{k}_2) \quad (76)$$

$$\mathcal{M}^\dagger = 2gg_2 \frac{\bar{v}_2 \gamma_5 u_1}{m^2 - s} (\vec{k}_1 \cdot \vec{k}_2) \quad (77)$$

2.2.2. $|\mathcal{M}|^2$

$$|\mathcal{M}|^2 = \mathcal{M}\mathcal{M}^\dagger \quad (78)$$

$$|\mathcal{M}|^2 = - \left(2gg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \right)^2 (\bar{u}_1 \gamma_5 v_2)(\bar{v}_2 \gamma_5 u_1) \quad (79)$$

2.2.3. *Spins*

I want the unpolarized cross section so I average over initial spins.

$$\mathcal{M} = \frac{1}{4} \sum_{s_1 s_2=1}^2 |\mathcal{M}|^2 \quad (80)$$

$$\mathcal{M} = -\frac{1}{4} \sum_{s_1 s_2=1}^2 \left(2gg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \right)^2 (\bar{u}_1 \gamma_5 v_2)(\bar{v}_2 \gamma_5 u_1) \quad (81)$$

$$\mathcal{M} = -\frac{1}{4} \left(2gg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \right)^2 \sum_{s_1 s_2=1}^2 (\bar{u}_1 \gamma_5 v_2)(\bar{v}_2 \gamma_5 u_1) \quad (82)$$

$$\mathcal{B} = \sum_{s_1 s_2=1}^2 (\bar{u}_1 \gamma_5 v_2)(\bar{v}_2 \gamma_5 u_1) \quad (83)$$

$$\mathcal{M} = -\frac{1}{4} \left(2gg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \right)^2 \mathcal{B} \quad (84)$$

$$\mathcal{B} = \sum_{s_1 s_2=1}^2 \bar{u}_1 \gamma_5 v_2 \bar{v}_2 \gamma_5 u_1 \quad (85)$$

$$\mathcal{B} = \sum_{s_1=1}^2 \bar{u}_1 \gamma_5 \left[\sum_{s_2=1}^2 v_{s_2}(\vec{p}_2) \bar{v}_{s_2}(\vec{p}_2) \right] \gamma_5 u_1 \quad (86)$$

$$\mathcal{B} = \sum_{s_1=1}^2 \bar{u}_1 \gamma_5 \left[\sum_{s=1}^2 v_s(\vec{p}_2) \bar{v}_s(\vec{p}_2) \right] \gamma_5 u_1 \quad (87)$$

$$\sum_{s=1}^2 v_s(\vec{p}) \bar{v}_s(\vec{p}) = \not{p} - M \quad (88)$$

$$\mathcal{B} = \sum_{s_1=1}^2 \bar{u}_1 \gamma_5 (\not{p}_2 - M) \gamma_5 u_1 \quad (89)$$

Like in HW 10 and my QED notes I introduce spinor component indices.

$$\mathcal{B} = \sum_{s_1=1}^2 (\bar{u}_1)_\alpha [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta (u_1)^\beta \quad (90)$$

I am using the Einstein summation convention to sum over spinor components.

$$\mathcal{B} = \sum_{s_1=1}^2 (u_1)^\beta (\bar{u}_1)_\alpha [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta \quad (91)$$

$$\mathcal{B} = \sum_{s_1=1}^2 (u_1 \bar{u}_1)_\alpha^\beta [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta \quad (92)$$

$$\mathcal{B} = \left[\sum_{s_1=1}^2 u_{s_1}(\vec{p}_1) \bar{u}_{s_1}(\vec{p}_1) \right]^\beta_\alpha [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta \quad (93)$$

$$\mathcal{B} = \left[\sum_{s=1}^2 u_s(\vec{p}_1) \bar{u}_s(\vec{p}_1) \right]^\beta_\alpha [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta \quad (94)$$

$$\sum_{s=1}^2 u_s(\vec{p}) \bar{u}_s(\vec{p}) = \not{p} + M \quad (95)$$

$$\mathcal{B} = (\not{p}_1 + M)_\alpha^\beta [\gamma_5 (\not{p}_2 - M) \gamma_5]^\alpha_\beta \quad (96)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) \gamma_5 (\not{p}_2 - M) \gamma_5 \right] \quad (97)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) (\gamma_5 \not{p}_2 \gamma_5 - M \gamma_5 \gamma_5) \right] \quad (98)$$

$$\gamma_5 \gamma^\mu = -\gamma^\mu \gamma_5 \implies \gamma_5 \not{p} = -\not{p} \gamma_5, \quad \gamma_5^2 = I \quad (99)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) (-\not{p}_2 \gamma_5 \gamma_5 - M \gamma_5^2) \right] \quad (100)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) (-\not{p}_2 \gamma_5^2 - M \gamma_5^2) \right] \quad (101)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) (-\not{p}_2 I - M I) \right] \quad (102)$$

I prefer to keep I implicit.

$$\mathcal{B} = \text{Tr} \left[(\not{p}_1 + M) (-\not{p}_2 - M) \right] \quad (103)$$

$$\mathcal{B} = -\text{Tr} \left[(\not{p}_1 + M) (\not{p}_2 + M) \right] \quad (104)$$

$$\mathcal{B} = -\text{Tr}(\not{p}_1 \not{p}_2 + M \not{p}_1 + M \not{p}_2 + M^2) \quad (105)$$

$$-\mathcal{B} = \text{Tr}(\not{p}_1 \not{p}_2) + M \cancel{\text{Tr}(\not{p}_1)} + M \cancel{\text{Tr}(\not{p}_2)} + M^2 \text{Tr}(I) \quad (106)$$

$$\text{Tr}(\not{p}) = p_\mu \text{Tr}(\gamma^\mu) = 0 \quad (107)$$

$$-\mathcal{B} = \text{Tr}(\not{p}_1 \not{p}_2) + M^2 \text{Tr}(I) \quad (108)$$

$$\mathcal{B} = -\text{Tr}(\not{p}_1 \not{p}_2) - 4M^2 \quad (109)$$

$$\mathcal{B} = -p_1^\mu p_2^\nu \text{Tr}(\gamma_\mu \gamma_\nu) - 4M^2 \quad (110)$$

$$\text{Tr}(\gamma_\mu \gamma_\nu) = 4\eta_{\mu\nu} \quad (111)$$

$\eta_{\mu\nu}$ is the Minkowski metric.

$$\mathcal{B} = -4p_1^\mu p_2^\nu \eta_{\mu\nu} - 4M^2 \quad (112)$$

$$\mathcal{B} = -4\vec{p}_1 \cdot \vec{p}_2 - 4M^2 \quad (113)$$

$$\mathcal{M} = \left(2gg_2 \frac{\vec{k}_1 \cdot \vec{k}_2}{s - m^2} \right)^2 (\vec{p}_1 \cdot \vec{p}_2 + M^2) \quad (114)$$

$$s = (\vec{k}_1 + \vec{k}_2)^2 = (\vec{p}_1 + \vec{p}_2)^2 \quad (115)$$

$$s = 2\lambda^2 + 2\vec{k}_1 \cdot \vec{k}_2 = 2M^2 + 2\vec{p}_1 \cdot \vec{p}_2 \quad (116)$$

$$\vec{k}_1 \cdot \vec{k}_2 = M^2 - \lambda^2 + \vec{p}_1 \cdot \vec{p}_2 \quad (117)$$

$$\boxed{p_{12} = \vec{p}_1 \cdot \vec{p}_2} \quad (118)$$

$$\boxed{\mathcal{M} = \left(2gg_2 \frac{M^2 - \lambda^2 + p_{12}}{2M^2 - m^2 + 2p_{12}} \right)^2 (p_{12} + M^2)} \quad (119)$$

2.2.4. Kinematics

I choose a frame where

$$\boxed{\vec{p}_1 = (E, 0, 0, p), \quad \vec{p}_2 = (E, 0, 0, -p), \quad E = \sqrt{M^2 + p^2}} \quad (120)$$

$$p_{12} = E^2 + p^2 = M^2 + p^2 + p^2 = M^2 + 2p^2 \quad (121)$$

$$\mathcal{M} = \left(2gg_2 \frac{2M^2 + 2p^2 - \lambda^2}{4M^2 + 4p^2 - m^2} \right)^2 (2M^2 + 2p^2) \quad (122)$$

$$\mathcal{M} = 2 \left(2gg_2 \frac{2M^2 + 2p^2 - \lambda^2}{4M^2 + 4p^2 - m^2} \right)^2 (M^2 + p^2) \quad (123)$$

$$\boxed{\mathcal{M} = 2 \left(2gg_2 \frac{2E^2 - \lambda^2}{4E^2 - m^2} \right)^2 E^2} \quad (124)$$

E is the initial energy of the electron and positron in the CM frame. Without loss of generality I orient my axes so that the final momenta are

$$\boxed{\vec{k}_1 = (E, k \sin \theta, 0, k \cos \theta), \quad \vec{k}_2 = (E, -k \sin \theta, 0, -k \cos \theta)} \quad (125)$$

where θ is the angle with the z (collision) axis, just like in spherical coordinates.

$$E^2 = k^2 + \lambda^2 \implies k^2 = E^2 - \lambda^2 \quad (126)$$

Since k is the magnitude of a 3-momentum: $k > 0$. Conservation of energy implies $E \geq \lambda$.

$$k = \sqrt{E^2 - \lambda^2} \quad (127)$$

$$p = \sqrt{E^2 - M^2} \quad (128)$$

2.2.5. Cross Section

I get the differential cross section in the CM frame from Eq. (5.32) in Schwartz.

$$\frac{d\sigma}{d\Omega} = \frac{\mathcal{M}}{64\pi^2 4E^2} \sqrt{\frac{E^2 - \lambda^2}{E^2 - M^2}} \quad (129)$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{128\pi^2} \sqrt{\frac{E^2 - \lambda^2}{E^2 - M^2}} \left(2gg_2 \frac{2E^2 - \lambda^2}{4E^2 - m^2} \right)^2 \quad (130)$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{1}{32} \left(\frac{gg_2}{\pi} \frac{2E^2 - \lambda^2}{4E^2 - m^2} \right)^2 \sqrt{\frac{E^2 - \lambda^2}{E^2 - M^2}}} \quad (131)$$